

Centralized Control & Monitoring System (CCMS)

A sustainable solution for smart cities of tomorrow



Centralized Control & Monitoring System

ViaLight's Centralized Control and Monitoring System (CCMS) offers a complete feeder control panel for a group of street lights. Typically, a group of 30-50 lighting poles are connected to the feeder panel, which is further connected to the Internet using a 4G network.

ViaLights has designed CCMS indigenously for Street Lighting Projects in India that control lighting schedules, monitor energy consumption and fault and theft detection. It applies to all types of luminaries, such as LEDs, HIDs, CFLs, FTLs, and induction lamps.



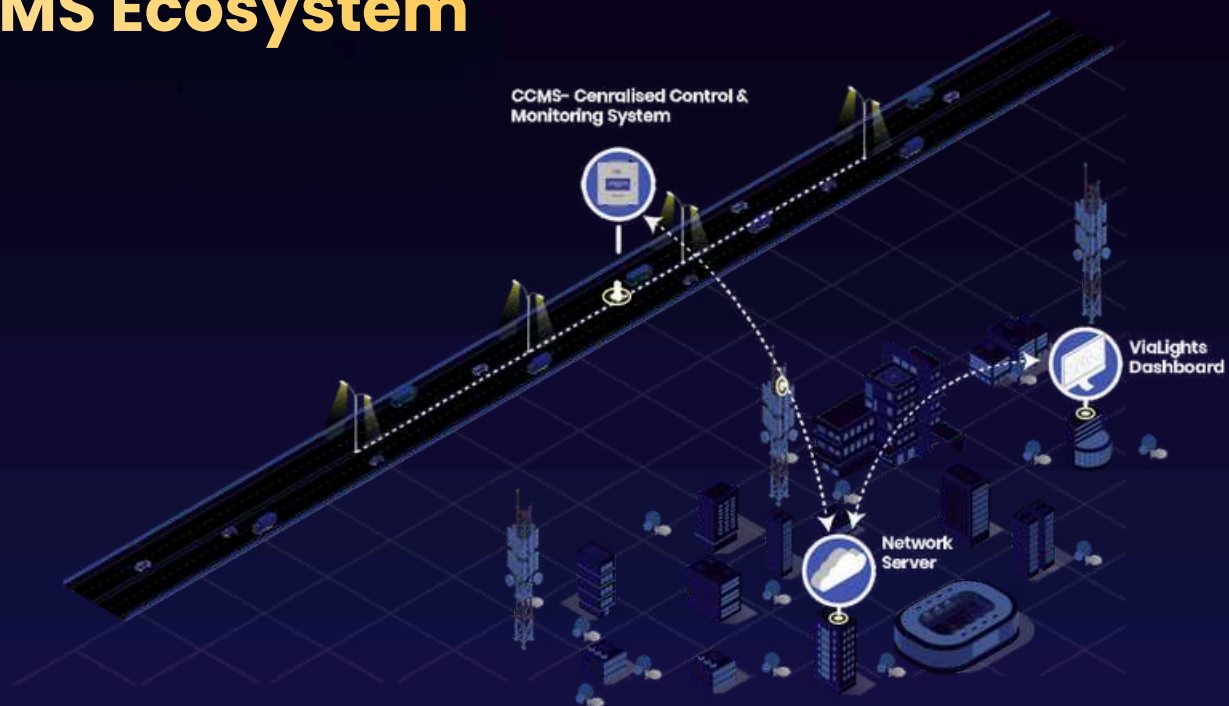
Features

- | | | | |
|--|---|--|-------------------------------------|
|  | Remote ON/OFF |  | Secure and tamper-proof technology |
|  | Offline data storage for up to 6 months |  | Secure communication |
|  | Astronomical / Independent schedule |  | Third party Web API integration |
|  | Four hours of RTU battery backup |  | Alert and Fault notification |
|  | Remote Management and Monitoring |  | Firmware update over the air (FOTA) |
|  | Fail safe with Bypass Switch |  | Over voltage load cut-off |
|  | 100 Years Calendar, RTC (Real Time Clock) with battery backup |  | Timer-Based Scheduling |

Alerts

- | | | | |
|--|-----------------------------------|--|---------------------------|
|  | Input / Output power failures |  | Meter communication fault |
|  | Contactor failure and Power theft |  | Earth leakage detection |
|  | High / Low voltage |  | Tilt and High temperature |
|  | Low power and Low power factor |  | Low battery |
|  | Door open | | |

CCMS Ecosystem



Smart Street Lighting Advantages



FOR CITIZENS



Improved public safety



Reduced Light pollution
and CO2 emission



FOR SMART CITIES



Eliminates electricity
power theft



Reduced time for repairs
and maintenance



FOR MANAGERS

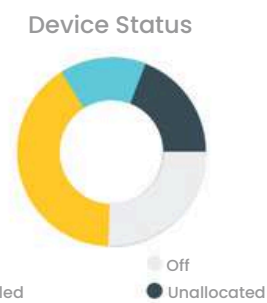
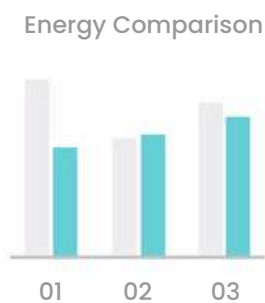


Remote access and monitoring of streetlight ecosystem



Reduced overall operational cost

CCMS Dashboard



Technical Specification

Hardware	
Processor	EC20CEFAG-512- SGNS
Antenna Options	External Antenna
Mounting Type	Single Door, Unibody Construction, Right Hand Door Hinge with Cam Lock, Foamed-On Door Gasket, Separate Component Mounting plate
IP Protection Rating	IP54 & IK08
Metering Measurement	Voltage, Current, Frequency, Power Factor, Power
Real Time Clock	Battery - Backup RTC
RTU Interfaces	One micro USB port. One extra relay Rs485 / RS232 for meter communication Ethernet port Extra DO's, DI's and I2c for sensor integration
Networking and Security	
Protocol	GSM / GPRS
Frequency Band	GPRS Quad band 850/900/1800/1900MHZ
Encryption	GPRS multi-slot class 12, coding scheme CS-1 to 4
Data Rate	Packet data up to 85.6K bps
Software and Control	
Notification	Input / Output power failures, Meter communication fault, Low power and Low power factor, High / Low voltage, Low battery, Tilt and High temperature
Monitoring Parameter	Voltage, Current, Power factor, Frequency, Power (kW), Energy (kWh)
Central Management System	Web-based software that allows remote configuration, scheduling, monitoring control and reporting
Electrical Specification	
Supply Type	Three Phase / Single Phase
Power Supply Range	90 VAC to 440 VAC, 50/60 Hz
Operating Temperature	-10°C to +55°C
Working Humidity Range	10% to 90% RH
Surge Protection	Type-2, 40 KA
Switching Load Capacity	40A per phase (Can modify as per requirement)
High Voltage Cutoff	290 VAC
Certificate	
Surge	Type-2, 40 KA
IP	IP54 & IK08
Energy Meter	Three/Single Phase IS 13779 Certification. Make: L&T / Secure / L&G / Genus Communication: RS 485/ RS232 Measurement accuracy: Class 1

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